EXPERIMENT 17:

AIM:

To write a C program to perform DES decryption by generating 16 round keys using the DES key-generation scheme and applying them in reverse order, from \(K_{16}\)

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
    char key[17], cipher[65];
    int i;

    printf("Enter 16-bit key (for demo): ");
    scanf("%16s", key);

    printf("Enter ciphertext: ");
    scanf("%64s", cipher);

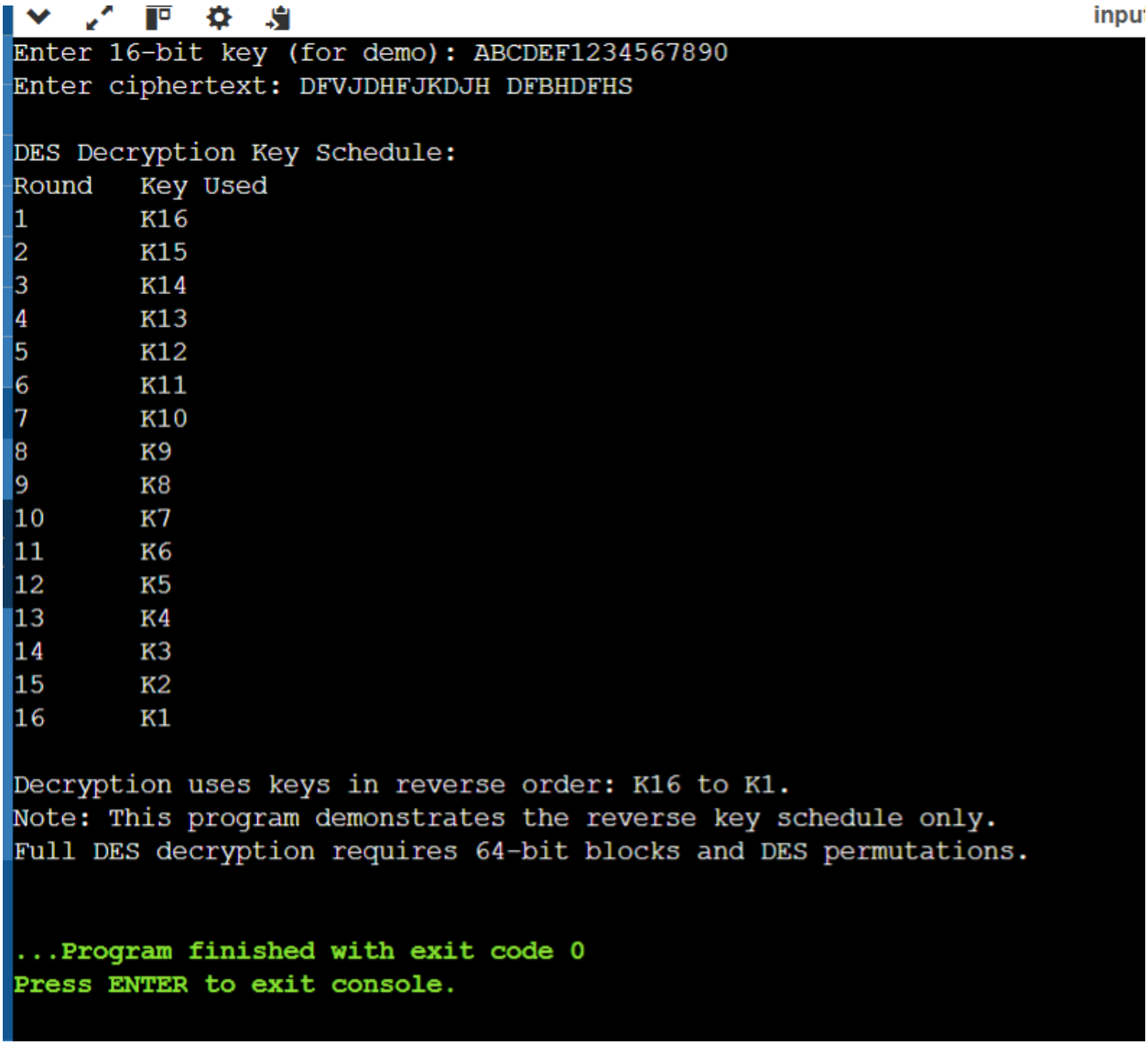
    printf("\nDES Decryption Key Schedule:\n");
    printf("Round\tKey Used\n");

    for (i = 16; i >= 1; i--)
        printf("%d\tK%d\n", 17 - i, i);
```

```
    printf("\nDecryption uses keys in reverse order: K16 to K1.\n");
    printf("Note: This program demonstrates the reverse key schedule only.\n");
    printf("Full DES decryption requires 64-bit blocks and DES permutations.\n");

    return 0;
}
```

OUTPUT:


inpu
Enter 16-bit key (for demo): ABCDEF1234567890
Enter ciphertext: DFVJDHFJKDJH DFBHDFHS
DES Decryption Key Schedule:
Round Key Used
1 K16
2 K15
3 K14
4 K13
5 K12
6 K11
7 K10
8 K9
9 K8
10 K7
11 K6
12 K5
13 K4
14 K3
15 K2
16 K1
Decryption uses keys in reverse order: K16 to K1.
Note: This program demonstrates the reverse key schedule only.
Full DES decryption requires 64-bit blocks and DES permutations.
...Program finished with exit code 0
Press ENTER to exit console.